The examination is being carried out on the following application documents

Description, Pages	1-31	as originally filed
Claims, Numbers	1-12	as originally filed
Drawings, Sheets	1/6-6/6	as originally filed

Reference is made to the following documents; the numbering will be adhered to in the rest of the procedure.

D1: WO 2022/030039 A1 (MMI SEMICONDUCTOR CO LTD [JP]) 10 February 2022

D2: WO 2021/078776 A1 (FLUSSO LTD [GB]) 29 April 2021

D3: US 10 094 691 B2 (SENSIRION AG [CH]) 9 October 2018

D4: JP 2002 156283 A (SEIKO EPSON CORP) 31 May 2002

D5: US 6 779 712 B2 (SENSIRION AG [CH]) 24 August 2004

The present Application does not meet the requirements of Article 52(1) EPC because the subject-matter of claim 1 does not meet the requirements of Article 54(3) EPC.

Furthermore, the subject-matter of claims 1, 2 and 8-12 does not meet the requirements of Article 56 EPC.

However, the subject-matter of claims 4-7 seems to meet the requirements of Article 56 EPC.

1 **Independent claim 1**

1.1 Article 54(3) EPC

1.1.1 Document **D1** discloses (the references between parentheses applying to this document):

A flow sensor chip (**fig. 4(a)**) comprising:

a substrate (**6 in fig. 4(a)**) in which a cavity (**10 in fig. 4(a)**) having an opening is formed (**as displayed in fig. 4(a)**);

a membrane (**5 in fig. 4(a)**) provided on a surface of the substrate so as to cover the opening (**as displayed in fig. 4(a)**);

a heater (**3 in fig. 4(a)**) provided inside the membrane (**5 in fig. 4(a)**);

a first thermopile (**7 on the left of 3 in fig. 4(a)**) and a second thermopile (**7 on the right of 3 in fig. 4(a)**) provided in the membrane (**as displayed in fig. 4(a)**),

the heater (**3 in fig. 4(a)**) being interposed between the first thermopile and the second thermopile (**as displayed in fig. 4(a)**); and

a heat conduction member (**12 in fig. 4(a)**) serving as a heat conduction path between the substrate and the membrane (**as displayed in fig. 4(a)**),

wherein a first hot junction (**8 of 7 on the left of 3 in fig. 4(a)**) and a first cold junction (**9 of 7 on the left of 3 in fig. 4(a)**) of a thermocouple provided in the first thermopile (**7 on the left of 3 in fig. 4(a)**) and a second hot junction (**8 of 7 on the right of 3 in fig. 4(a)**) and a second cold junction (**9 of 7 on the right of 3 in fig. 4(a)**) of a thermocouple provided in the second thermopile (**7 on the right of 3 in fig. 4(a)**) are arranged at positions overlapping with the opening as viewed from a direction normal to the surface of the substrate (**overlapping of 7 with the opening of 10 as displayed in fig. 4a**), and

the heat conduction member includes:

a first heat conduction member (**12 on the left in fig. 4(a)**) serving as a heat conduction path between the first cold junction (**9 in fig. 4(a)**) and the substrate (**6 in fig.4(a)**); and

a second heat conduction member (**12 on the right in fig. 4(a)**) serving as a heat conduction path between the second cold junction (**9 in fig. 4(a)**) and the substrate (**6 in fig.4(a)**).

Hence, the subject-matter of **independent claim 1** does not meet the requirements of Article 54(3) EPC.

1.1.2 Document **D2** also discloses (the references between parentheses applying to this document):

A flow sensor chip (**figs. 5 and 6A-C**) comprising:

a substrate (**1 in figs. 5 and 6A-C**) in which a cavity having an opening is formed (**as displayed in figs. 5 and 6A-C**);

a membrane (**2 in figs. 5 and 6A-C**) provided on a surface of the substrate so as to cover the opening (**as displayed in figs. 5 and 6A-C**);

a heater (**3 in figs. 5 and 6A-C**) provided inside the membrane (**as displayed in fig. 5**);

a first thermopile (**6a in figs. 5 and 6A-C**) and a second thermopile (**6b in figs. 5 and 6A-C**) provided in the membrane (**as displayed in figs. 5 and 6A-C**),

the heater (**3 in figs. 5 and 6A-C**) being interposed between the first thermopile and the second thermopile (**as displayed in figs. 5 and 6A-C**); and

a heat conduction member (**5 in fig. 6A**) serving as a heat conduction path between the substrate and the membrane (**as displayed in fig. 6A**),

wherein a first hot junction and a first cold junction of a thermocouple provided in the first thermopile (**6a in figs. 5 and 6A-C**) and a second hot junction and a second cold junction of a thermocouple provided in the second thermopile (**6b in figs. 5 and 6A-C**) are arranged at positions overlapping with the opening as viewed from a direction normal to the surface of the substrate (**as displayed in fig. 6A the positions overlap the opening**), and

the heat conduction member includes:

a first heat conduction member (**5 in fig. 6A**) serving as a heat conduction path between the first cold junction and the substrate (**1 in fig. 5**); and

a second heat conduction member (**other 5 in fig. 6A**) serving as a heat conduction path between the second cold junction and the substrate (**1 in fig. 5**).

Hence, the subject-matter of **independent claim 1** does not meet the requirements of Article 54(3) EPC.

1.2 Article 56 EPC

1.2.1 Document **D3** is considered as the closest prior art of the subject-matter of **independent claim 1**.

D3 discloses (the references between parentheses applying to this document):

A flow sensor chip (**fig. 4**) comprising:

a substrate (**1 in fig. 4**) in which a cavity having an opening is formed (**opening delimited by the dashed line in fig. 4**);

a membrane (**3 in fig. 4**) provided on a surface of the substrate so as to cover the opening (**dashed line in fig. 4**);

a heater (**4 in fig. 4**) provided inside the membrane (**col. 7 lines 8-9: "the following sensor elements are all arranged on or in the membrane: A heater 4..."**);

a first thermopile (**7 in fig. 4**) and a second thermopile (**6 in fig. 4**) provided in the membrane (**3 in fig. 4**),

the heater (**4 in fig. 4**) being interposed between the first thermopile (**7 in fig. 4**) and the second thermopile (**6 in fig. 4**); and

~~a heat conduction member serving as a heat conduction path between the substrate and the membrane,~~

wherein a first hot junction (**73 in fig. 4**) and a first cold junction (**79 in fig. 4**) of a thermocouple provided in the first thermopile (**7 in fig. 4**) and a second hot junction (**63 in fig. 4**) and a second cold junction (**69 in fig. 4**) of a thermocouple (**6 in fig. 4**) provided in the second thermopile (**6 in fig. 4**) are arranged at positions overlapping with the opening as viewed from a direction normal to the surface of the substrate (**as displayed in fig. 4**), and

~~the heat conduction member includes:~~

~~a first heat conduction member serving as a heat conduction path between the first cold junction and the substrate; and~~

~~a second heat conduction member serving as a heat conduction path between the second cold junction and the substrate.~~

The subject-matter of **independent claim 1** differs from the known flow sensor chip in that it comprises:

a heat conduction member serving as a heat conduction path between the substrate and the membrane,

wherein the heat conduction member includes:

a first heat conduction member serving as a heat conduction path between the first cold junction and the substrate; and

a second heat conduction member serving as a heat conduction path between the second cold junction and the substrate.

1.2.2 The technical effect associated to this difference is that the cold junction is protected and the conduction member contributes uniformizing the temperature of the cold junction with the frame.

1.2.3 The objective technical problem may be seen as how to improve the sensor of **D3** so that it is mechanically more robust and the cold reference is more stable.

1.2.4 Document **D4** discloses a sensor with thermopiles where a cold junction is protected with an infrared shielding member made of metal (**16 in fig. 1, paragraph [0017]**). This shielding member improves the conduction performance of the sensor and helps having a uniform temperature of the cold junction (**paragraph [0019]**).

Document **D5** discloses a sensor with thermopiles where a cold junction is protected from undesired mechanical stress with a solder bump made of metal that is also impose per conduction the cold temperature as the one of the environment far from the heater (**col. 4 lines 56-59, col. 3 lines 40-46**).

In front of the objective technical problem and starting from **D3**, the person skilled in the art would combine the teaching of **D4** or of **D5** with the sensor of **D3** obtaining the subject-matter of **independent claim 1**.

Therefore, the subject-matter of **independent claim 1** does not meet the requirements of Article 56 EPC.

2 **Dependent claims 2 and 8-12**

2.1 Article 56 EPC

2.1.1 Document **D3** is also considered as the closest prior art of the following **claims:**

Claim 2: The person skilled in art, combining the teaching of **D4** or **D5** with the sensor of **D3**, would obtain the subject-matter of **claim 2** as the configuration of the conducting members is the one of **D4** or **D5**.

Claim 8: The person skilled in the art, combining the teaching of **D5** with the sensor of **D3**, would obtain the subject-matter of **claim 8** as the conduction member would be in contact with the ground potential in **D5**.

Claim 9: The person skilled in art, combining the teaching of **D4** or **D5** with the sensor of **D3**, would obtain the subject-matter of **claim 9** as the material in **D4** would be any metal **D4** (**paragraph [0017]**) and copper in **D5** (**col. 3 line 65 - col. 4 line 9**).

Claims 10-11: For the person skilled in the art, the design details related to a metal terminal would be implemented according the configuration without involving inventive skills.

Claim 12: For the person skilled in the art, a rectangular shape conduction member would be obtained by combining the teaching of **D4** or of **D5** with the sensor of **D3**.

3 **Dependent claims 3-7**

3.1 Article 56 EPC

3.1.1 Document **D3** is also considered as the closest prior art of the following **claims:**

The subject-matter of **claim 3** seems to meet the requirements of Article 56 EPC as the teaching of **D4** or **D5** is to cover the cold junction and not having a conduction member extending from the substrate to a side of the cold junction.

The subject-matter of **claim 4** is depending on **claim 3**. Therefore the subject-matter of **claim 4** seems also to meet the requirements of Article 56 EPC.

When the subject-matter of **claim 5** is depending on **claim 3**, the subject-matter of **claim 5** seems also to meet the requirements of Article 56 EPC.

When the subject-matter of **claim 5** is depending on **claim 2**, the subject-matter of **claim 5** seems also to meet the requirements of Article 56 EPC because **D4** or **D5** do not hints towards a solution where the conduction member and the membrane would be perforated.

The subject-matter of **claim 6** is depending on **claim 5**. Therefore the subject-matter of **claim 6** seems also to meet the requirements of Article 56 EPC.

The subject-matter of **claim 7** is depending on **claim 5**. Therefore the subject-matter of **claim 7** seems also to meet the requirements of Article 56 EPC.

4 Additional remarks

4.1 Article 84 EPC

4.1.1 The application does not meet the requirements of Article 84 EPC, because **claim 1** is not clear.

Indeed following a common definition of "overlapping": "covering something partly by going over its edge, or covering part of the same space" (https://dictionary.cambridge.org/fr/dictionnaire/anglais/overlapping).

It is clear in **D1** that the thermopiles have junctions overlapping the cavity opening as the thermopiles are partly going over the edge of the cavity.

However, it seems that the Application discloses support only for the second meaning where "overlapping" relates to covering part of the same space.

It is suggested to clarify the meaning of overlapping with:

"wherein the first cold junction and the second cold junction are not included in the area where the substrate and the membrane are in contact with each other, but are included in an area where the membrane faces the cavity, not facing the substrate," with bases in the description (**paragraph [0009]**).

5 Final remarks

5.1 The objections addressed against **claim 1** may conceivably be overcome if **claim 1** was drafted with the features found in **claims 3, 4, 5, 6** or **7** and clarified according to **paragraph [0009]**.

5.2 Should the applicant regard this particular matter as patentable, an independent claim should be filed taking account of Rule 43(1) EPC. The applicant should also indicate how the subject-matter of the new claim differs from the state of the art and the significance thereof.

5.3 When filing amended claims the applicant should at the same time bring:

- the independent claims in the two-part form in accordance with Rule 43(1) EPC, with those features known in combination from the prior art being placed in the preamble (Rule 43(1)(a) EPC) and the remaining features being included in the characterising part (Rule 43(1)(b) EPC).

- the features of the claims should be provided with reference signs placed in parentheses to increase the intelligibility of the claims (Rule 43(7) EPC). This applies to both the preamble and characterising portion.

- the description in conformity with the amended claims and **D3** should be identified in the description and the relevant background art disclosed therein should be briefly discussed (Rule 42(1)(b)).

5.4 The attention of the applicant is drawn to the fact that the application may not be amended in such a way that it contains subject-matter which extends beyond the content of the application as filed (Article 123(2) EPC).

5.5 In order to facilitate the examination of the conformity of the amended application with the requirements of Article 123(2) EPC, the applicant should clearly identify the amendments made, irrespective of whether they concern amendments by addition, replacement or deletion, and indicate the passages of the application as filed on which these amendments are based (see Guidelines H-III, 2.2). If the applicant considers it appropriate, these indications could be submitted in handwritten form on a copy of the relevant parts of the application as filed or in documents with tracked changes.